

MANIKAVASAGAM S

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PROFILE SUMMARY

Computer Science graduate with hands-on experience in Java, Spring Boot, Microservices, REST APIs, and distributed backend systems. Built scalable backend applications using Spring Boot, Spring Data JPA, Hibernate, Kafka, Redis, PostgreSQL, and AWS. Strong foundation in Object-Oriented Programming, Data Structures Algorithms, SQL, microservices architecture, and backend system design, with experience developing, testing, debugging, and optimizing backend applications.

TECHNICAL SKILLS

- **Languages:** Java, JavaScript, TypeScript, SQL
- **Backend:** Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, API Design, JWT, Web-Socket, OpenFeign, WebClient, Maven
- **Microservices / Architecture:** Microservices, API Gateway, Eureka, Resilience4J, Saga Pattern, Outbox/Inbox Pattern, Event-Driven Architecture
- **Databases:** PostgreSQL, MySQL, MongoDB, Qdrant
- **Messaging & Caching:** Redis, Apache Kafka
- **DevOps / Cloud / Monitoring:** Docker, Kubernetes, AWS (EC2, S3, RDS, IAM, Lambda, CloudFront, EKS, ECS), CI/CD, Prometheus, Grafana, OpenTelemetry
- **Testing / CS Fundamentals:** JUnit 5, Mockito, Data Structures & Algorithms, OOP, SOLID Principles, Design Patterns, SDLC, OS, Networks, Git, GitHub, Postman
- **AI-ML:** Spring AI, LLMs, Prompt Engineering, RAG, Embeddings, LangChain, LangGraph, Hugging Face (Transformers), AI Agents, MCP
- **Frontend:** React.js, HTML, CSS

PROJECTS

Distributed Fintech Wallet System | Java, Spring Boot, Microservices, REST APIs, Kafka, Redis, PostgreSQL

- Built a distributed wallet and transaction platform using Java, Spring Boot, and Microservices, developing secure REST APIs for wallet management, payments, and transaction processing with PostgreSQL, Redis, and Kafka.
- Implemented Saga-based distributed transaction workflows with Outbox/Inbox patterns and idempotent Kafka consumers to maintain consistency across microservices and prevent duplicate transaction processing.
- Reduced database reads by 95% using Redis write-through caching (@Cacheable) for frequently accessed wallet and transaction data, improving response time and lowering database IOPS
- Improved concurrency handling and prevented conflicting updates using Redis distributed locks (SETNX), transactional workflows, and partition-aware Kafka message processing
- Increased system throughput by 3x-5x and reduced end-to-end latency by up to 80% by introducing asynchronous event-driven processing for transaction, notification, and wallet update workflows
- Added observability and distributed tracing with Prometheus, Grafana, and OpenTelemetry to monitor service health, latency, and microservice interactions

Real-Time Meeting Collaboration System | Spring Boot, WebSocket, React, MongoDB

- Built a real-time AI-powered video conferencing and collaborative note-taking platform using React, Spring Boot, WebSocket-based communication, and Python, supporting live captions, chat, and AI-generated meeting summaries
- Implemented low-latency transcript streaming using the Web Speech API and WebSockets to deliver real-time speech-to-text updates during meetings

- Developed secure backend modules for JWT authentication, room-based session management, participant moderation, and real-time chat synchronization
- Built a Python-based NLP pipeline for extractive summarization, keyword-based action-item detection, and concise meeting summary generation from transcript data
- Designed event-driven transcript handling and persistent storage using MongoDB and WebSocket-based streaming to support post-meeting summary generation and collaborative review workflows

Movie Ticket Booking System | React, Spring Boot, MySQL, WebSocket, Elasticsearch

- Developed a full-stack movie ticket booking platform using React and Spring Boot with JWT-based authentication, role-based access control, and modular REST APIs for users, admins, and theatre owners
- Designed a real-time seat locking mechanism using WebSockets and soft locks with expiration, handling concurrent bookings and preventing race conditions during seat selection
- Implemented transactional booking workflows and concurrency-safe seat allocation logic to ensure atomic booking confirmation under simultaneous user requests
- Integrated Elasticsearch to optimize movie, theatre, and show search performance, offloading high-volume search queries from the primary relational database
- Added API rate limiting using Bucket4j (token bucket algorithm) to protect backend services from request spikes and improve service reliability under heavy traffic

INTERNSHIP EXPERIENCE

UpEd Learning Solutions | Frontend Web Development Intern | May 2026

- Contributed to the development of NexusFlow, a project management platform, using React and modern web development practices
- Developed and improved frontend pages and reusable UI components for project and task management features
- Worked on responsive and user-friendly interfaces for different screen sizes
- Integrated frontend functionality with application APIs and handled user interactions and data display
- Debugged UI issues, fixed functionality problems, and performed testing to improve application usability

EDUCATION

B.E Computer Science Engineering
KLN College of Engineering (2022 - 2026)

CGPA: 7.5 / 10

ACHIEVEMENTS & CERTIFICATIONS

- Solved 200+ Data Structures and Algorithms problems on LeetCode
- Java Backend Development – GeeksforGeeks
- Full Stack Development – upGrad
- Java Programming – GeeksforGeeks